

ALEXEY GALLEGO

Systems Builder | Product Development | Software Engineering | Business Systems
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SUMMARY

Throughout my career, I have repeatedly been trusted with problems I had never solved before and technologies I had never used before. Whether developing industrial automation systems, leading ERP initiatives, building business software, mentoring engineers, or adopting emerging AI workflows, I have built a reputation for learning quickly, taking ownership, and delivering results. My passion is helping organizations grow by turning complex operational challenges into scalable systems, products, and processes.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, Node.js, React, Next.js, Angular, ASP.NET, Python, Java, HTML, CSS, Tailwind CSS

AI & Automation: LLM Engineering, Agentic AI, MCP, OpenAI APIs, Claude APIs, AI-Assisted Development, Prompt Engineering, Retrieval-Augmented Generation (RAG), Codex, Claude Code

Databases: SQL Server, PostgreSQL, MySQL, MongoDB

Infrastructure & DevOps: Docker, Kubernetes, Linux, CI/CD, GitHub Actions, AWS, Nginx, Ubuntu Server, VMware

Methodologies: Agile/Scrum, Technical Leadership, Code Reviews, Mentorship.

EXPERIENCE

Systems & Software Engineer | Encore Automation | Auburn Hills, MI

Jul 2021 – Present

- Identified operational inefficiencies affecting more than 100 employees and architected a company-wide business operations platform from scratch (Node.js, React, SQL), replacing spreadsheet-driven processes with automated timesheets, expense management, scheduling, approval workflows, and reporting that improved visibility and accountability across the organization.
- Increased timesheet compliance from approximately 70% to 95% and expense submission compliance from approximately 50% to 90% by redesigning user workflows, implementing automated notifications, and creating manager dashboards that eliminated manual administrative follow-up.
- Led technical architecture and infrastructure decisions for internal business applications, maintaining 99%+ uptime through ownership of Ubuntu-based production environments, Dockerized deployments, CI/CD pipelines, and application monitoring.
- Strengthened company security posture by replacing legacy authentication approaches with modern JWT-based authentication, refresh token rotation, CSRF protection, secure password management, and role-based access controls across multiple business systems.
- Rebuilt the company's digital presence using Next.js, modernizing product marketing, improving mobile usability, increasing online visibility, and creating a stronger platform for customer engagement and lead generation.
- Rebuilt the company's web presence from the ground up using Next.js, replacing an outdated site with outdated product information, and extended brand reach by producing and editing video content for LinkedIn and YouTube.
- Directed the technical transformation of a company-wide ERP initiative, leading two engineers while defining architecture, security, infrastructure, development standards, and deployment strategy across engineering, purchasing, finance, receiving, and project management operations.
- Introduced AI-assisted engineering workflows and intelligent receipt processing capabilities that reduced administrative review effort from approximately one day to one to two hours while accelerating software delivery and improving operational scalability.
- Mentored multiple engineering interns across software and automation projects, providing technical guidance, architecture reviews, and hands-on support while fostering engineering growth and development.
- Led development of a patent-pending robotic sealing solution deployed across automotive manufacturing facilities in the United States and Canada, reducing cycle time by approximately 50%, improving product quality, reducing material waste, and creating a high-margin product that generated significant revenue for the business.

PROJECTS

AI Receipt Analysis System

- Designed and implemented an AI-powered expense processing pipeline using OpenAI Vision that analyzes receipt images, extracts structured line-item data, validates submissions, and automatically matches expenses to company records. Integrated as a core ERP module, reducing administrative review effort from approximately one business day to one to two hours while improving accuracy and scalability.

Digital Twin Website - (Portfolio: <https://alex-portfolio-psi.vercel.app/en>)

- Designed and developed a personal portfolio website showcasing engineering, software development, automation, AI, and product development projects. Created to communicate technical expertise, leadership experience, and business impact through project case studies, architecture decisions, and measurable outcomes while serving as a central hub for professional branding and career opportunities.

Industrial Process Validation System

- Developed a Python-based industrial quality monitoring platform that interfaces directly with FANUC robots, captures real-time process telemetry, automatically identifies defective components, and provides production visibility dashboards. Improved process reliability from approximately 80% to 99% while reducing manufacturing risk and improving quality assurance.

Legal Operations SaaS Platform (Ongoing)

- Building a SaaS platform that transforms manual client intake and onboarding processes into scalable digital workflows for law firms and professional-service organizations. Responsible for product vision, technical architecture, and full-stack development, leveraging AI, workflow automation, and modern web technologies to help firms improve operational efficiency, reduce administrative burden, and gain greater visibility into client onboarding and case preparation.

EDUCATION

B.S. Mechanical Engineering | Florida International University, Miami, FL | May 2021